Subject Code 25ES1EE101	R25

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

B.Tech. I Year I Semester Regular Examinations, January 2026

BASIC ELECTRICAL ENGINEERING

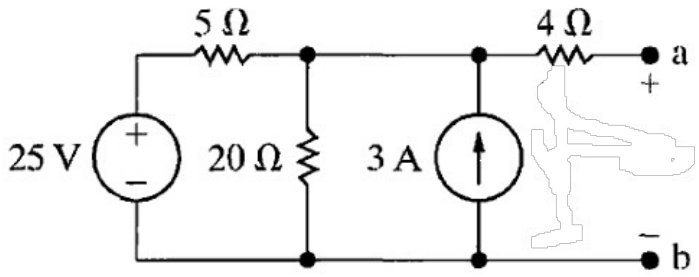
(Common to CSE and IT)

Time: 3 hours **Max. Marks: 60**

Answer ALL questions in PART-A.
Answer any ONE question from each unit in PART-B.

a)	State Superposition theorem and mention one application of it.	2M	CO-1	BL-1
b)	Differentiate between real power and reactive power in AC circuits.	2M	CO-2	BL-2
c)	What is the significance of back EMF in a DC motor?	2M	CO-3	BL-2
d)	Define slip and slip speed of a three-phase induction motor.	2M	CO-3	BL-1
e)	List any two advantages of a lead-acid battery.	2M	CO-4	BL-2

	UNIT-I			
1	a) Explain the role of electrical energy in modern life and various engineering branches.	5M	CO-1	BL-2
	b) Explain Kirchhoff's laws with suitable examples.	5M	CO-1	BL-3
	OR			
2	State and explain Thevenin's theorem and apply it to solve for the circuit given below between terminals a and b.	10 M	CO-1	BL-4

				
UNIT-II				
3	Analyze a series RLC circuit under sinusoidal excitation and derive expressions for current and impedance.	10 M	CO-2	BL-4
OR				
4	Explain three-phase balanced circuits and derive voltage and current relations for star and delta connections.	10 M	CO-2	BL-3
UNIT-III				
5	a) Explain the working principle of a transformer.	5M	CO-3	BL-2
	b) Derive the EMF equation of a DC generator from fundamentals.	5M	CO-3	BL-3
OR				
6	Explain the equivalent circuit of a practical transformer and derive expressions for efficiency and regulation.	10 M	CO-3	BL-4
UNIT-IV				
7	a) Explain the principle of operation of a three-phase induction motor.	5M	CO-3	BL-2
	b) Explain torque–slip characteristics of an induction motor.	5M	CO-3	BL-4
OR				
8	Explain voltage regulation of a synchronous generator using synchronous impedance method.	10 M	CO-3	BL-3
UNIT-V				
9	a) Explain the basic layout of a domestic electrical installation.	5M	CO-4	BL-2

	b) Explain the need for earthing and types of earthing.	5M	CO-4	BL-2
OR				
10	Explain the construction, working and characteristics of a lead-acid battery and explain UPS operation with block diagram.	10M	CO-4	BL-3

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